CodeGhar

Your Code. Every Version. Always Recoverable.

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Project Overview

CodeGhar is a lightweight, web-based version control system for text and source code files. It allows users to upload, create, edit, and track multiple versions of files with an intuitive interface. The system provides a GitHub-like edit view, version restation, and secure file management without complex Git commands.

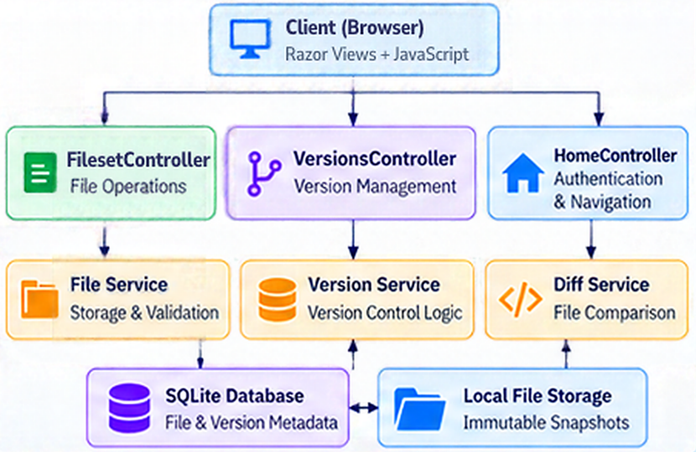
Key Features

- **Upload & Create** — Add text and code files with strict validation
- **Restore Versions** — Revert to any previous version without losing history
- **Edit & Save** — Automatic version creation on every change
- **Code Diff Viewer** — Line-by-line comparison with highlighted changes
- **Version History** — Track all versions with imestamps and change summary
- **Secure & Fast** — Local storage, file validation and optimized performance

Technology Stack

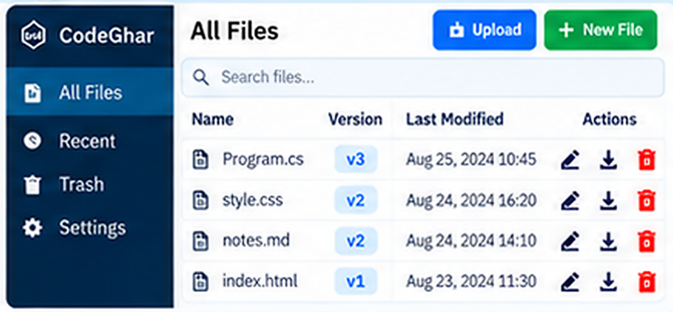
- ASPNET Core MVC (C#)
- Razor Views
- SQLite Database
- HTML5, CSS3, JavaScript
- Monaco Editor (Code Editor)
- Myers Diff Algorithm
- Bootstrap (UI Components)

System Architecture

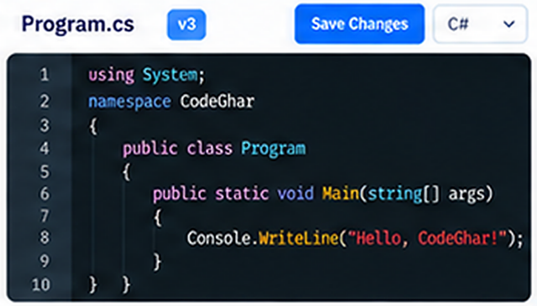


Application Screenshots

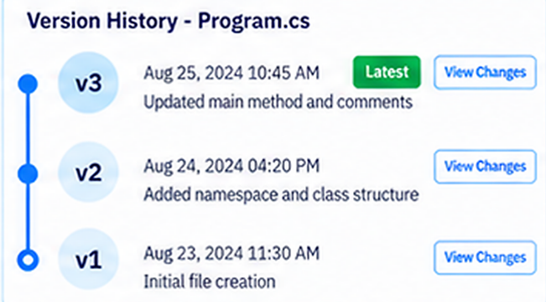
1 File Dashboard



2 Code Editor



3 Version History



4 Code Diff Viewer

Supported File Types

Text Files	.txt, .md, .markdown
Web Files	.html, .css, .js, .ts, .jsx, .tsx
Source Code	.cs, .java, .py, .c, .cpp, .php, .go
Config Files	.json, .xml, .yaml, .yml, .gitignore
Others	.sql, .sql, .bat, .ps1, .rs, .vue

Version Lifecycle

v1 File Created → v2 Edited & Saved → v3 More Changes → v4 Restore Previous Version

Restoring a version creates a new version without deleting existing history.

Security & Validation

- Only text and code files allowed
- File size limit (2 MB)
- Strict file extension validation
- Path traversal protection
- Files stored outside wwwroot
- User authentication and authorization
- Antiforgery protection for all actions

Use Cases

- Manage and track code snippets
- Maintain notes and documentation
- Keep versioned configuration files
- Collaborate in academic projects
- Learn version control concepts easily

Project Impact

- Simplifies version control for students
- Easy-to-use, intuitive interface
- No need for complex Git commands
- Helps in learning and managing code history
- Useful for academic and personal projects
- Lightweight and runs locally